

Load Control

1 Core Idea

Definition

Load control is the operating system's decision to reduce the number of processes competing for memory when the system is thrashing.

Even with a good page replacement algorithm and good global frame allocation, the system can still thrash if total memory demand is too high.

Thrashing Condition

$$\sum \text{working sets of runnable processes} > \text{available physical memory}$$

2 Why Replacement Alone Is Not Enough

Page replacement decides which page to evict. Load control decides whether too many processes are active at the same time.

Mechanism	Question answered
Page replacement	Which page should be removed?
Frame allocation	How many frames should each process receive?
Load control	How many processes should compete for memory now?

Key Problem

If every active process needs more memory and no process can safely give memory away, page replacement cannot solve the problem.

3 PFF Symptom

The **Page Fault Frequency (PFF)** algorithm can reveal memory overload.

Thrashing Signal

Thrashing is likely when PFF says some processes need more frames, but no processes have a low enough fault rate to give frames back.

In this situation, giving more frames to one process would hurt another process. The system needs to reduce the active load.

4 Main Solution: Swap Out Processes

Load Reduction

The OS can swap one or more processes to disk, freeing all page frames held by those processes.

The freed frames are then given to the remaining active processes.

1. Detect that the system is thrashing.
2. Pick a process to swap out.
3. Write its pages and process state to disk as needed.
4. Free its page frames.
5. Redistribute those frames among remaining processes.
6. If thrashing continues, swap out another process.

5 Why Swapping Still Matters with Paging

Paging moves individual pages. Swapping removes an entire process from memory.

Technique	What it moves	Purpose here
Paging	Individual pages.	Handle normal page faults.
Swapping	Whole processes or most of their resident pages.	Reduce total memory demand.

Practical Meaning

With virtual memory, swapping is still useful, but mainly as a load-control tool.

6 Connection to Two-Level Scheduling

Load control is similar to two-level scheduling.

Level	Role
Longer-term decision	Which processes stay in memory?
Short-term scheduler	Which in-memory runnable process gets the CPU next?

The OS may periodically swap some processes in and others out, keeping just enough active processes to make the page-fault rate acceptable.

7 Degree of Multiprogramming

Definition

The **degree of multiprogramming** is the number of processes kept active in main memory.

Load control must avoid two extremes.

Extreme	Problem
Too many active processes	Combined working sets exceed memory, causing thrashing.
Too few active processes	CPU may sit idle while processes wait for I/O.

8 Choosing Which Process to Swap Out

The OS should consider more than just process size.

Factor	Why it matters
Process size	Swapping out a large process frees more frames.
Page fault rate	A process causing many faults may be worsening thrashing.
CPU-bound or I/O-bound	Removing too many CPU-bound or I/O-bound processes can unbalance the system.
Remaining process mix	The OS needs enough runnable work to keep the CPU busy.

Tradeoff

Swapping out processes reduces memory pressure, but swapping out too many processes can reduce CPU utilization.

9 Load Control Flow

Decision Flow

high page fault rate
→ PFF shows processes need more frames
→ no process can give frames back
→ swap out one or more processes
→ redistribute freed frames

10 Final Summary

One-Box Summary

Load control handles thrashing caused by too many active processes.
If combined working sets exceed RAM, replacement algorithms are not enough.
The OS can swap out processes to free frames for the rest.
The goal is enough processes for CPU use, but not so many that memory thrashes.